

Regular Article

Association of Protein and Endotoxin in Outdoor Air with Emergency Department Visits for Children and Adults with Asthma in Fukuoka, Japan

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We examined the association of biological components in airborne particles, *i.e.*, proteins and endotoxins, in outdoor air with asthma exacerbation in the Fukuoka metropolitan area, Fukuoka, Japan. Data on emergency department (ED) visits for asthma in children (age, 0–14 years) and adults (age, 15–64 years) were collected at a medical center from December 2014 to November 2015. One hundred eighty-one children and 143 adults visited the ED for asthma, and the weekly number of ED visits in children increased in autumn, *i.e.*, September (second week) to November (first week). Fine (aerodynamic diameter $\leq 2.5\mu\text{m}$) and coarse ($\geq 2.5\mu\text{m}$) particles were collected for 3 or 4 weeks per month, and protein and endotoxin concentrations were analyzed. Protein was largely prevalent in fine particles ($0.34\text{--}7.33\mu\text{g}/\text{m}^3$), and concentrations were high in April, May, June, and October. In contrast, endotoxin was mainly included in coarse particles ($0.0010\text{--}0.0246\text{EU}/\text{m}^3$), and concentrations were high in September (third week), October (first, second, and fourth weeks), February (fourth week), and July (first week). The results of a Poisson regression analysis indicated that endotoxin (in fine and coarse particles alike) was a significant factor for ED visits related to asthma in children, even after adjusting for meteorological factors, *i.e.*, temperature, relative humidity, and wind speed. However, there was no association between environmental factors and ED visits for asthma in adults. These results suggest that endotoxin in outdoor air is significantly associated with an increased risk of asthma exacerbation in children.

Key words asthma attack; bioaerosol; lipopolysaccharide; air pollution; fine particle; coarse particle

INTRODUCTION

Asthma is the most common respiratory disease in many countries, including Japan.^{1–3} It is characterized by chronic inflammation of lung airways and bronchial hyperresponsiveness. Variable air-flow obstruction, or an asthma attack, is caused by various stimuli.^{4,5} Kanchongkittiphon *et al.*⁶ have reviewed publications from 2000 to 2013 assessing the relationship between indoor exposure to environmental factors and asthma symptoms to update prior conclusions on the topic as presented by the Committee on the Assessment of Asthma and Indoor Air of the Institute of Medicine.⁷ They have concluded that there is indeed sufficient evidence on the association between indoor exposure to various allergens and endotoxins and exacerbation of asthma.⁶

Proteins are typical allergens, and many studies have reported that proteins are detected in suspended particles in outdoor air.^{8–12} Endotoxins are major components of the outer membrane of Gram-negative bacteria, and the inhalation of

endotoxins is known to induce inflammation in respiratory organs.¹³ Endotoxins are also detected in suspended particles in outdoor air,^{14–17} and Yoda *et al.* have reported that endotoxin concentration in indoor air is significantly correlated with that in outdoor air.¹⁸

These reports suggest that protein and endotoxin in outdoor air may affect asthmatic symptoms, and it is important to understand the effect of these substances on asthma exacerbation to avoid the burden on the health caused by it. However, there are only a few reports on the association between such atmospheric substances and asthma exacerbation. In our previous study, we reported that the concentration of endotoxins in airborne particles in outdoor air was significantly and positively associated with the number of emergency department (ED) visits for asthma in Kyoto, Japan.¹⁹ Although the sensitivity to protein and endotoxin may vary according to the patient's age, age was not considered in the previous study. Moreover, it is important to assess the relationship between protein and endotoxin in outdoor air with asthma exacerbation in other districts

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that have distinct environmental conditions to understand the effect of environmental factors on asthmatic symptoms.

The purpose of the present study was to examine the association of asthma with protein and endotoxin in outdoor air in the Fukuoka metropolitan area, which includes Fukuoka City and surrounding satellite cities in Fukuoka Prefecture, located about 510 km west of Kyoto City. Kyoto is a part of an inland basin, the Kyoto Basin, whereas Fukuoka City faces the Sea of Japan and is relatively warmer in winter. In this study, we assessed the association of atmospheric protein and endotoxin with asthma by considering patient age, *i.e.*, children and adults, and meteorological factors such as temperature, relative humidity, and wind speed, which have been reported to be associated with asthma exacerbation.^{20,21)}

MATERIALS AND METHODS

Patient Data Anonymous unlinked data on ED visits for asthma were collected at Fukuoka Tokushukai Medical Center (33.543772 degrees north latitude, 130.447182 east longitude) in the Fukuoka metropolitan area from December 2014 to November 2015. The data included information on sex and age of the patient, the date of consultation, main complaint, diagnosis, and outcome. The subjects of this study were children aged 0–14 years and adults aged 15–64 years. The total number of children and adults who visited ED for asthma was counted. This study was approved by the ethics committee of Kyoto Pharmaceutical University (Approval No. 19-16-16) and Fukuoka Tokushukai Medical Center (Approval No. 280801).

Analysis of Protein and Endotoxin To analyze protein and endotoxin, we collected fine (aerodynamic diameter $\leq 2.5 \mu\text{m}$) and coarse particles (aerodynamic diameter $\geq 2.5 \mu\text{m}$) in the Fukuoka metropolitan area at the Fukuoka Institute of Health and Environmental Sciences (33.512399 degrees north latitude, 130.500192 east longitude). The monitoring station was located approximately 7.5 km south of the medical center. Fine and coarse particles were collected on quartz and glass filters, respectively, at a flow rate of $1 \text{ m}^3/\text{min}$ for 1 week per filter using a high-volume air sampler (HV1000R, Shibata Scientific Technology, Soka, Japan) equipped with an impactor (code: 080130-067, Shibata Scientific Technology) installed on the roof of the four-floor monitoring station, approximately 18 m high. The impactor has a size selective inlet, and fine and coarse particles are trapped on quartz and glass filters, respectively. Collection was performed from December 2014 to November 2015, and 84 samples (42 sets of fine and coarse particles, respectively) were collected (Table 1). Before collection, the filters were heated to 250°C for 2 h. After collection, the filters were kept at -80°C until sample solutions were prepared to analyze the protein and endotoxin.

Protein and endotoxin were analyzed using methods described previously.¹²⁾ In brief, 15% of the sample filters (corresponding to 1512 m^3 of air) were extracted with 0.025% Tween 20 *via* sonication for 30 min. After centrifugation of the extracts, a portion of the supernatant was used to analyze the protein and endotoxin.

Proteins and endotoxins were analyzed using a bicinchoninic acid assay (Micro BCA™ Protein Assay Kit, Thermo Scientific, Waltham, MA, U.S.A.) and the kinetic chromogenic limulus amoebocyte lysate method (Limulus Color KY Test Wako Kit, FUJIFILM Wako Pure Chemical Corporation,

Table 1. Periods of Sample Collection

Year	Month	Week
2014	December	1st (December 1–7), 2nd (8–14), 3rd (15–21)
2015	January	1st (January 6–12), 2nd (13–19), 3rd (20–26)
	February	1st (February 2–8), 2nd (9–15), 3rd (16–22), 4th (February 23–March 1)
	March	1st (March 2–8), 2nd (9–15), 3rd (16–22), 4th (23–29)
	April	1st (April 2–8), 2nd (9–15), 3rd (16–22), 4th (23–29)
	May	1st (May 7–13), 2nd (14–20), 3rd (21–27)
	June	1st (June 1–7), 2nd (15–21), 3rd (22–28)
	July	1st (July 3–9), 2nd (10–16), 3rd (17–23), 4th (24–30)
	August	1st (August 3–9), 2nd (10–16), 3rd (17–23)
	September	1st (September 4–10), 2nd (11–17), 3rd (September 25–October 1)
	October	1st (October 2–8), 2nd (9–15), 3rd (16–22), 4th (23–29)
	November	1st (November 4–10), 2nd (11–17), 3rd (18–24), 4th (November 25–December 1)

Osaka, Japan), respectively, using a microplate reader (Sunrise Thermo RC-R). Both assays were performed according to the manufacturer's protocols. Concentrations of protein and endotoxin in all 84 samples were higher than the detection limits (protein $2.5 \mu\text{g}/\text{mL}$ and endotoxin $0.0005 \text{ EU}/\text{mL}$). For a blank filter, the concentrations of protein and endotoxin were lower than the detection limit.

Temperature, Relative Humidity, and Wind Speed Weekly mean values of temperature, relative humidity, and wind speed were calculated from their daily mean values obtained from the Japan Meteorological Agency.²²⁾

Statistical Analysis We used Microsoft Excel 2016 to calculate the correlation coefficients. We applied a Poisson generalized linear model to analyze the association between the weekly numbers of ED visits for asthma and the weekly levels of environmental factors. The analyses were performed using SPSS Statistics Version 22 (IBM Software Group, Chicago, IL, U.S.A.). *p*-Values of less than 0.05 were considered to indicate statistical significance.

RESULTS

ED Visits for Asthma Figure 1 shows the number of ED visits for asthma in children (0–14 years of age) and adults (15–64 years of age), a total of 181 and 143, respectively, during the study period. In children, the weekly number of ED visits ranged from 0 to 16, increasing greatly in autumn, *i.e.*, from the second week of September to the first week of November. In contrast, the variation in the number of weekly ED visits by adults was not large, *i.e.*, only from 0 to 8.

Atmospheric Protein, Endotoxin, and Meteorological Factors Figure 2 shows the concentrations of proteins and endotoxins in fine and coarse particles during the study period. In most weeks, the concentration of protein in fine particles ($0.34\text{--}7.33 \mu\text{g}/\text{m}^3$, the mean value $\pm$ standard deviation $3.28 \pm 1.66 \mu\text{g}/\text{m}^3$) was higher than that in coarse particles ($0.04\text{--}1.41$, $0.62 \pm 0.32 \mu\text{g}/\text{m}^3$), especially during the spring months, including April (third and fourth weeks) and May (third week), and occasionally in June (first week), and October (fourth week). In contrast, the endotoxin concentration was higher in coarse particles ($0.0010\text{--}0.0246$, $0.0063 \pm 0.0045 \text{ EU}/\text{m}^3$) than in fine particles ($0.0001\text{--}0.0132$,

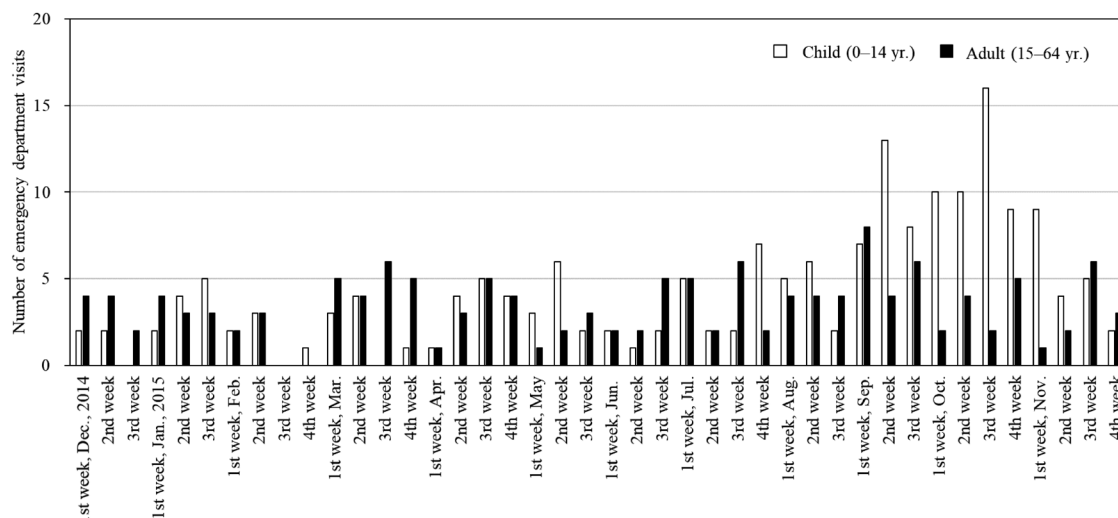


Fig. 1. Number of Emergency Department Visits by Children and Adults for Asthma

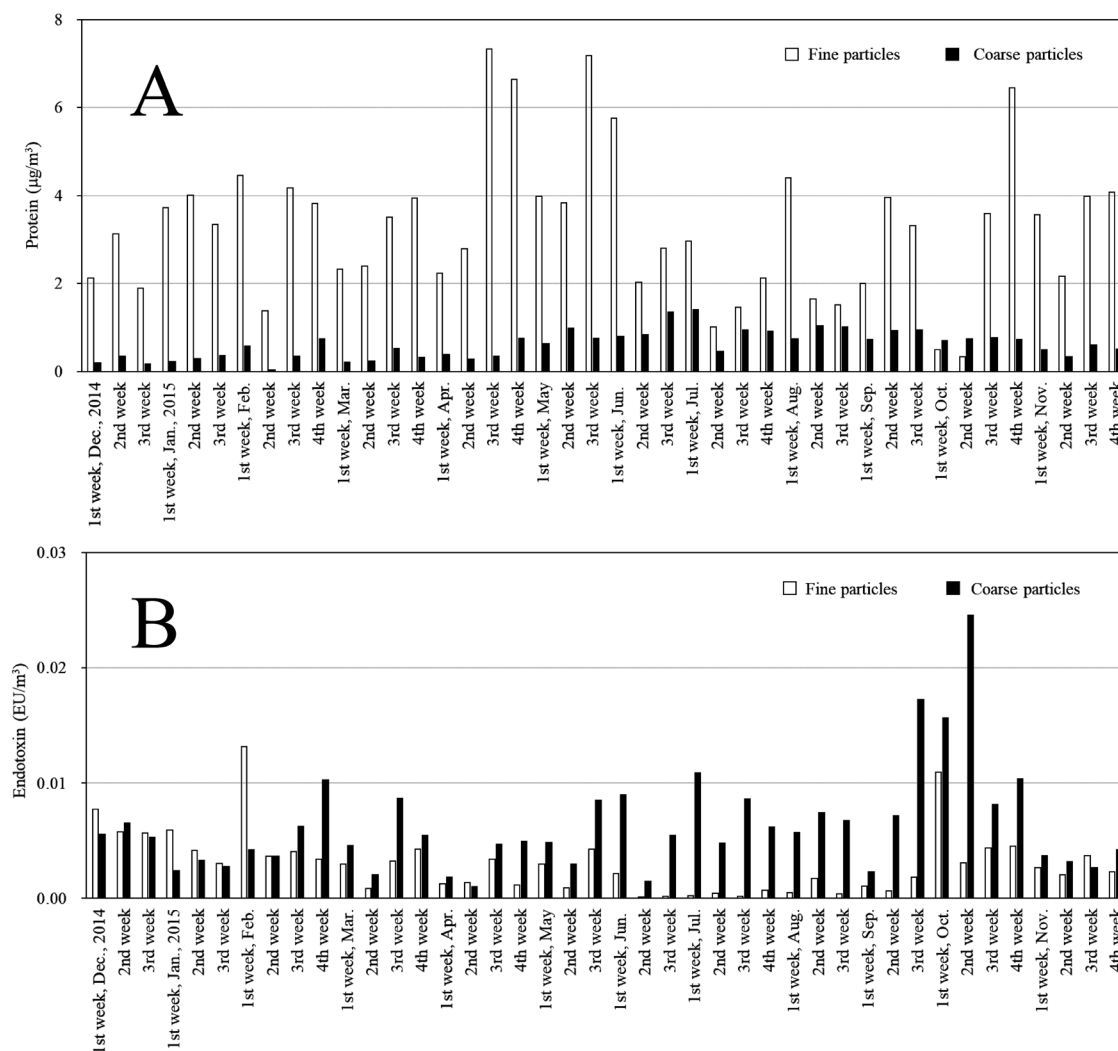


Fig. 2. Concentrations of Protein (A) and Endotoxin (B) in Fine and Coarse Particles in Outdoor Air

$0.0030 \pm 0.0028 \text{ EU}/\text{m}^3$) and especially high during autumn months, including September (third week) and October (first, second, and fourth weeks), and also sporadically in February (fourth week), and July (first week).

Table 2 shows the monthly values of meteorological factors,

i.e., temperature, relative humidity, and wind speed (weekly data are shown in Supplementary Table 1). The monthly temperature was high in July (26.1°C), August (28.2°C), and September (23.0°C). Similarly, the monthly relative humidity was high in June (79.8%), July (81.6%), and September (78.1%).

The monthly wind speed did not show large fluctuations during the study period.

Table 3 shows the correlation coefficients for atmospheric

Table 2. Monthly Values of Meteorological Factors Including Temperature, Relative Humidity, and Wind Speed

Period	Temperature (°C)	Relative humidity (%)	Wind speed (m/s)
December 2014	7.4	58.5	4.0
January 2015	8.4	61.5	3.1
February	7.7	62.2	3.0
March	10.8	59.8	3.0
April	16.1	69.0	3.2
May	20.6	68.0	2.7
June	22.4	79.8	2.7
July	26.1	81.6	2.7
August	28.2	75.7	2.6
September	23.0	78.1	3.4
October	19.0	66.5	2.6
November	16.1	76.1	2.5
Entire Period	17.0	69.6	2.9

protein, endotoxin, temperature, relative humidity, and wind speed. The concentration of endotoxin in fine particles was negatively correlated with protein in coarse particles, temperature, and relative humidity. The concentration of protein in coarse particles was positively correlated with endotoxin in coarse particles, temperature, and relative humidity. In addition, the concentrations of protein and endotoxin in coarse particles were negatively correlated with wind speed. All correlations observed in this study were moderate or weak.

Association of Protein and Endotoxin in Outdoor Air with ED Visits for Asthma The association of the concentrations of protein and endotoxin in fine and coarse particles with the number of ED visits for asthma in children and adults was assessed by a Poisson regression model to adjust for meteorological factors. As shown in Table 4, endotoxin concentrations in fine and coarse particles were statistically significant factors in ED visits for children with asthma. Temperature was a statistically significant factor in the number of ED visits for asthma in children in all cases. In contrast, none of the studied environmental factors were statistically significant in determining the number of ED visits for adults.

Table 3. Correlation Coefficients and Corresponding *p*-Values for Protein, Endotoxin, and Meteorological Factors

	Endotoxin (fine particle)	Protein (coarse particles)	Endotoxin (coarse particles)	Temperature	Relative humidity	Wind speed
Protein (fine particle)	0.079 (0.620)	0.018 (0.910)	-0.142 (0.369)	-0.072 (0.649)	-0.266 (0.088)	-0.194 (0.219)
Endotoxin (fine particle)	1.000	-0.385 (0.012)	0.142 (0.371)	-0.560 (<0.001)	-0.501 (0.001)	0.194 (0.219)
Protein (coarse particles)		1.000	0.392 (0.010)	0.763 (<0.001)	0.576 (<0.001)	-0.502 (0.001)
Endotoxin (coarse particles)			1.000	0.25 (0.111)	-0.007 (0.966)	-0.302 (0.052)
Temperature				1.000	0.615 (<0.001)	-0.363 (0.018)
Relative humidity					1.000	-0.255 (0.152)

Correlation coefficient was calculated with the weekly values of protein, endotoxin, temperature, relative humidity, and wind speed. Statistically significant: *p* < 0.05.

Table 4. Poisson Regressions of Protein and Endotoxin with Meteorological Factors on Emergency Department Visits for Asthma

Variables	Children			Adults		
	Coef	S.E.	<i>p</i> -Value	Coef	S.E.	<i>p</i> -Value
Protein (fine particles) ($\mu\text{g}/\text{m}^3$)	-0.034	0.047	0.476	0.021	0.054	0.700
Intercept	2.484	0.862	0.004	0.389	0.901	0.666
Temperature (°C)	0.040	0.014	0.003	0.023	0.016	0.143
Relative humidity (%)	-0.009	0.009	0.301	-0.003	0.011	0.760
Wind speed (m/s)	-0.339	0.157	0.031	0.202	0.149	0.175
Endotoxin (fine particles) (EU/m^3)	71.96	30.46	0.018	-30.25	42.14	0.473
Intercept	1.309	0.838	0.118	0.887	0.843	0.293
Temperature (°C)	0.051	0.015	<0.001	0.019	0.017	0.268
Relative humidity (%)	-0.002	0.009	0.862	-0.006	0.010	0.531
Wind speed (m/s)	-0.308	0.154	0.045	0.187	0.144	0.195
Protein (coarse particles) ($\mu\text{g}/\text{m}^3$)	0.010	0.388	0.979	0.495	0.456	0.277
Intercept	2.145	0.731	0.003	0.480	0.747	0.521
Temperature (°C)	0.039	0.016	0.016	0.011	0.019	0.556
Relative humidity (%)	-0.007	0.090	0.420	-0.007	0.011	0.475
Wind speed (m/s)	-0.313	0.165	0.058	0.258	0.158	0.101
Endotoxin (coarse particles) (EU/m^3)	38.90	14.48	0.007	14.26	19.05	0.454
Intercept	1.448	0.784	0.065	0.354	0.811	0.662
Temperature (°C)	0.033	0.014	0.020	0.020	0.016	0.207
Relative humidity (%)	-0.003	0.009	0.750	-0.003	0.103	0.774
Wind speed	-0.230	0.155	0.140	0.214	0.148	0.149

Coef: regression coefficient, S.E.: standard error of the regression. Statistically significant: *p* < 0.05.

DISCUSSION

We examined the association of protein and endotoxin in outdoor air with ED visits for asthma in children and adults in the Fukuoka metropolitan area, and found that the concentrations of endotoxin in fine and coarse particles were significantly and positively associated with the number of ED visits for asthma in children after adjustment for meteorological factors. In contrast, no significant association was found between the concentrations of protein and endotoxin and ED visits for asthma in adults. To the best of our knowledge, this is the first report to have considered the differences in response to atmospheric protein and endotoxin in terms of age.

Toxicologic and epidemiologic studies have demonstrated that inhaled endotoxin induces airway inflammation *via* Toll-like receptor 4 and is an established risk factor for asthma.^{6,23)} Rabinovitch *et al.* measured exposures of asthmatic children (aged 6–13 years) to endotoxin in airborne particles and found that increase in endotoxin exposures were associated with increased asthmatic symptoms and decreased values of forced expiratory volume in 1 s.²⁴⁾ This information supports our findings that atmospheric endotoxin levels are associated with ED visits for asthma in children.

In the present study, outdoor endotoxin was a significantly positive factor in ED visits for asthma in children, but not in adults, suggesting that children are more susceptible to the effect of endotoxins. Zhang *et al.* conducted a systematic review and meta-analysis to obtain the overall risk ratios of daily hospital utilization for asthma associated with air pollutants in East Asia and showed that concentrations of nitrogen dioxide, ozone, and particulate matter with aerodynamic diameters less than 10 and 2.5 μm (PM_{10} and $\text{PM}_{2.5}$) were significantly and positively associated with the risk of hospital admissions in children (aged 0–14 years) but not adults (aged 15–64 years).²⁵⁾ Several physiological characteristics of children have been highlighted to be the cause of such high vulnerability to air pollution; their lungs and immune systems are not fully developed, they breathe more air per unit body weight and are typically more active than adults, and their peripheral airways are smaller than that of adults.^{26,27)} These features may be the cause for the observed differences in the association of outdoor endotoxin concentration with ED visits for asthma in children and adults.

Protein concentrations were not associated with the number of ED visits for asthma in children or adults in the present study. Song *et al.* reported that aerosol proteinaceous matter was mainly derived from agricultural activities, biomass burning, and fugitive dust or soil resuspension.¹¹⁾ In a previous study, we found that protein concentration was positively associated with the concentrations of organic material combustion in Sasebo, Nagasaki, Japan.¹²⁾ The protein detected in the present study may have originated from such combustion.

This study has several limitations. First, the concentrations of proteins and endotoxins in outdoor air measured at a monitoring station close to the medical center where we collected patient data may include some errors with respect to the personal exposure levels because of the substances originated from indoor sources. However, concentrations of proteins and endotoxins in outdoor air could be used as a proxy to presume personal exposure levels because endotoxin concentration in indoor air is reported to be significantly correlated with that

in outdoor air.¹⁸⁾ Second, we used ED visit as a surrogate for asthma exacerbation. There is a time lag between exposure to increased levels of protein and endotoxin and ED visits to treat asthma exacerbation, and the variations in time lag among patients may affect the statistical association between the concentration of protein and endotoxin and ED visits. Third, this study used data on ED visits from only one medical center. This has resulted in limited statistical power to detect differences among subgroups of children for statistical analysis. Finally, we used weekly levels of protein and endotoxin because daily measurements of these substances were too laborious for a long-term study. Further studies using daily data should be performed in the future.

In this study, we found that endotoxin in outdoor air was a significant factor in the increase of ED visits for asthma in children, but not in adults. This is the first report on the varying effects of age on the association of atmospheric protein and endotoxin with asthma exacerbation. This study shows the importance of studying biological substances in outdoor air to understand the causes of asthma exacerbation.

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Conflict of Interest The authors declare no conflict of interest.

Supplementary Materials The online version of this article contains supplementary materials.

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